Consumer Complaint Trends and Response Intelligence

AnalystLab Africa • Data Analytics Internship • Project Report

Project	Consumer Complaint Trends and Response Intelligence Platform
Dataset	62,516 Bank of America complaint records from the CFPB, covering 2017 to 2023
Role	Data Analyst Intern, AnalystLab Africa
Tools	Power BI (visualization), Power Query with M code (data preparation), DAX (KPI calculations)

1. Project README and Data Quality Documentation

Overview

This project analyses consumer complaint volumes, product mix, response performance and complaint-forwarding speed for Bank of America, using complaint records published by the Consumer Financial Protection Bureau (CFPB). The data was prepared in Power Query, the KPIs were built in DAX, and the findings are presented in an interactive Power BI dashboard.

Data cleaning and validation

Check	Finding	Action taken
Uniqueness	62,516 unique Complaint IDs and 0 duplicates.	No records were removed.
Timely response? (blank)	1,494 blank values, about 2.4% of all records.	Replaced with “In progress” instead of “No”, as the project instructions required. This reflects active status and does not distort the timely response metric.
Public response (blank)	2,175 blank values, about 3.5% of all records.	Kept as blank. A blank means no public response was recorded, so nothing was filled in.

Engineered features

- **Forwarding Interval:** the date difference between “Date submitted” (to the CFPB) and “Date received” (by the company).
- **Year, Month and Quarter:** extracted from the submission date to support time-series analysis.

2. Analytical Scripts and KPI Definitions

All measures are written in DAX against the `'Data'` table.

Total Complaints

The number of distinct complaints in the current filter context.

```
Total Complaints = DISTINCTCOUNT('Data'[Complaint ID])
```

Complaints In Progress

The number of complaints whose timely response status is “In progress”.

```
Complaints In Progress =
CALCULATE([Total Complaints], 'Data'[Timely response?] = "In progress")
```

Average and Median Forwarding Interval

The mean and the median number of days between submission and receipt. The median is shown beside the average because a few very late complaints can pull the average upward.

```
Avg Forwarding Interval = AVERAGE('Data'[forwarding_interval])
Median Forwarding Interval = MEDIAN('Data'[forwarding_interval])
```

Timely Response %

The share of complaints answered on time. Records marked “In progress” are excluded so that the denominator contains only valid Yes or No responses.

```
Timely Response % =
VAR ValidResponses =
    CALCULATE(
        [Total Complaints],
        'Data'[Timely response?] IN { "Yes", "No" }
    )
VAR TimelyCount =
    CALCULATE(
        [Total Complaints],
        'Data'[Timely response?] = "Yes"
    )
RETURN
    DIVIDE(TimelyCount, ValidResponses, 0)
```

% Contribution by Product

Each product’s share of all complaints. `ALL('Data')` removes the product filter so the denominator is the full total.

```
% Contribution by Product =
DIVIDE(
    [Total Complaints],
    CALCULATE([Total Complaints], ALL('Data'))
)
```

YoY Volume Change %

The percentage change in complaint volume compared with the previous year. Variables are used to find the previous year’s complaints based on the submission year, and the percentage difference is then calculated from the two values.

```
Previous Year Complaints =
VAR CurrentYear = SELECTEDVALUE('Data'[Year])
RETURN
    CALCULATE([Total Complaints], 'Data'[Year] = CurrentYear - 1)

YoY Volume Change % =
DIVIDE(
    [Total Complaints] - [Previous Year Complaints],
    [Previous Year Complaints],
    0
)
```

3. Business Insights and Recommendations

Five key insights

1. **Volumes rose sharply from 2022.** Complaints surged starting in 2022 and peaked in 2023, a clear break from the fairly stable levels between 2017 and 2020.
2. **Two products dominate.** “Checking or savings accounts” and “Credit reporting” are the most frequently reported products and account for a large share of the total complaint volume.

3. **Response performance is strong.** The timely response rate is about 96%, which points to good compliance with response-time standards for the first handling of complaints.
4. **Forwarding is very fast.** The average forwarding interval is about 1.2 days, and most complaints are forwarded on the day they are submitted (0 days).
5. **A few complaints are forwarded late.** Some outliers took more than 15 days to be forwarded, mostly linked to Checking or savings accounts and Mortgage products.

Three recommendations

1. **Investigate the 2022 to 2023 surge.** Run a root-cause review to find out whether it comes from new product launches, systemic banking issues or wider economic factors.
2. **Shift resources to the main problem products.** Direct more customer service and quality assurance effort to Checking or savings accounts and Credit reporting, which are the main sources of consumer friction.
3. **Track the "In progress" complaints.** Set up internal tracking for the 1,494 complaints still marked "In progress", so each one is resolved and none goes on to miss the timely response threshold.

Points to keep in mind

- The timely response rate of about 96% leaves out the 1,494 "In progress" complaints, so it describes only complaints with a known Yes or No outcome.
- Blank timely response values were treated as "In progress" under the project instructions.